Uch miqdor nisbati

Ikki nisbatda bitta miqdor takrorlansa, ulardan uch miqdorning nisbatini yozish mumkin.

1-QISM

Mavzu



Bularni avvalgi darslarda o'rganganmiz.

ESLAB OLAYLIK

- Nisbatning **ikkala** hadini bitta songa ko'paytirsak, teng kuchli nisbat hosil bo'ladi: $3:7=9:21$ (×3).
- Ikkala hadni bitta songa bo'lsak ham teng kuchli nisbat chiqadi: $10:4=5:2$. Boshqa qisqarmasa — qisqarmas ko'rinish.
- Ikki sonning dastlabki umumiy karralisi (qisqacha EKUK) — ikkalasiga ham bo'linadigan eng kichik son: 4 va 6 niki — 12, 3 va 5 niki — 15.
- A soni B ning $\frac{2}{5}$ qismi → $A:B=2:5$. A soni B dan 3 marta ko'p → $3:1$, 3 marta kam → $1:3$. A soni B ning yarmi → $1:2$.

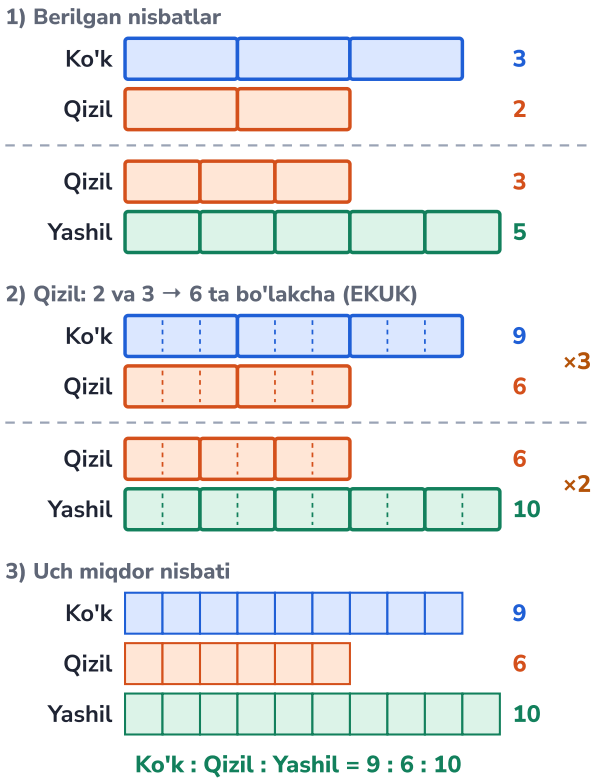
QOIDA · UCH MIQDOR NISBATINI TOPISH

$X:Y$ va $Y:Z$ nisbatlar berilgan. Y ikkala nisbatda ham qatnashgan.

- Taqqoslash uchun Y ni har ikkala nisbatda ham bir xil bo‘lakchalar orqali tasvirlaymiz.
- Y ga mos ikki sonning dastlabki umumiy karralisini topamiz (masalan, 4 va 5 ning dastlabki umumiy karralisi 20) va har bir nisbatning ikkala hadini mos songa ko'paytiramiz.
- Y ikkala nisbatda ham bir xil bo‘laklar orqali tasvirlangani sababli $X:Y:Z$ ni yozamiz.

1-MISOL · BAYROQCHALAR

Gulchambarda ko'k : qizil bayroqchalar $= 3:2$, qizil : yashil bayroqchalar $= 3:5$. Ko'k : qizil : yashil nisbatini toping.



1. Qizil birinchi nisbatda 2 bo'lak, ikkinchisida 3 bo'lak. 2 va 3 ning EKUKi — 6.
2. $3:2 = 9:6$ (×3), $3:5 = 6:10$ (×2).
3. Ko'k : qizil : yashil $= 9:6:10$.

2-MISOL · SHARLAR

Oq : pushti sharlar $= 3:2$, oq : binafsha sharlar $= 5:4$. Oq : pushti : binafsha nisbatini toping.

1. Bu yerda oq ikkala nisbatda birinchi turibdi: 3 va 5. EKUK — 15.
2. $3:2 = 15:10$ (×5), $5:4 = 15:12$ (×3).
3. Oq : pushti : binafsha $= 15:10:12$.



MIQDORLARNI TOPISH

Ma'lum son necha bo'lak ekanini aniqlaymiz. Keyin 1 bo'lakni, so'ng kerakli miqdorni topamiz:

$$\text{3 bo'lak} = 42 \;\rightarrow\; \text{1 bo'lak} = 42 : 3 = 14 \;\rightarrow\; \text{10 bo'lak} = 10 \cdot 14 = 140$$

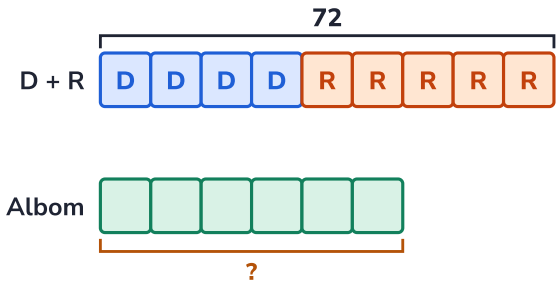
1-MISOL · BAYROQCHALAR SONI

Gulchambarda ko'k : qizil : yashil bayroqchalar $= 9 : 6 : 10$. Qizil bayroqchalar 24 ta. Hammasi bo'lib nechta bayroqcha bor?

1. Qizil — 6 bo'lak: 6 bo'lak = 24.
2. 1 bo'lak $= 24 : 6 = 4$.
3. Hammasi $9 + 6 + 10 = 25$ bo'lak: $25 \cdot 4 = 100$ ta.

2-MISOL · DO'KONDAGI BUYUMLAR

Do'konda daftar : ruchka : albom $= 4 : 5 : 6$. Daftar va ruchkalar birgalikda 72 ta. Albomlar nechta?



D — daftarlar (4 bo'lak), R — ruchkalar (5 bo'lak)

1. Daftar va ruchka: $4 + 5 = 9$ bo'lak = 72.
2. 1 bo'lak $= 72 : 9 = 8$.
3. Albomlar: $6 \cdot 8 = 48$ ta.

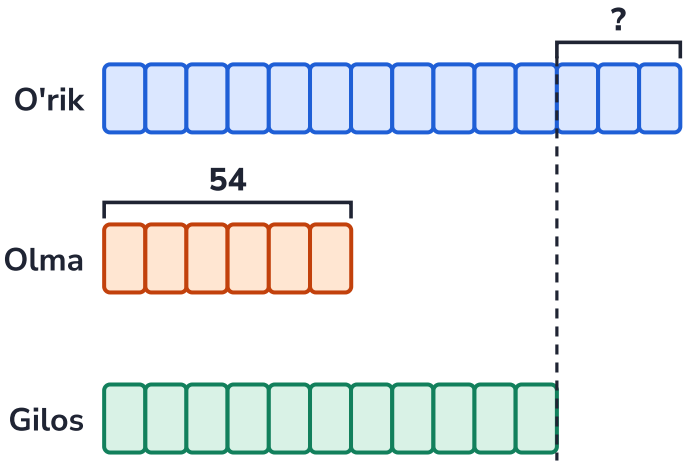
2-QISM

Misollar ishlash

1-MISOL · BOG'

Bog'da o'rik : olma daraxtlari $= 7:3$. Olma daraxtlari gilos daraxtlarining $\frac{6}{11}$ qismi. Olma daraxtlari 54 tup. a) O'rik, olma va gilos daraxtlari nisbatini toping. b) O'rik daraxtlari gilosdan necha tup ko'p?

1. Olma : gilos $= 6:11$. *(Olma gilosning $\frac{6}{11}$ qismi.)*
2. $7:3 = 14:6$. *(Olmaga mos sonlar 3 va 6, EKUK 6.)*
3. a) O'rik : olma : gilos $= 14:6:11$.



4. 6 bo'lak = 54, 1 bo'lak $= 54:6 = 9$.
5. b) O'rik gilosdan $14 - 11 = 3$ bo'lak ko'p: $3 \cdot 9 = 27$ tup. *(O'rik 126 tup, gilos 99 tup.)*

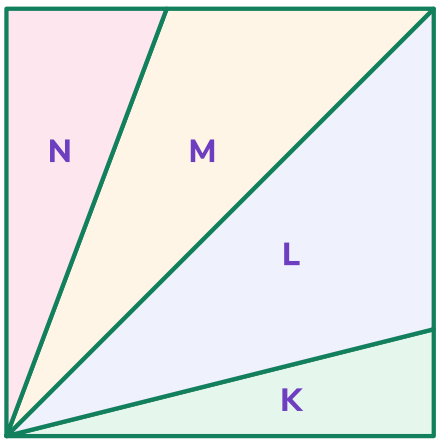
2-MISOL · MATOLAR

Tikuvchida uch xil mato bor. Atlas : adras $= 3:5$, adras : shoyi $= 3:2$ (uzunliklari). Atlas adrasdan 18 m qisqa. a) Atlas : adras : shoyi nisbatini toping. b) Adras shoyidan necha metr uzun? c) Uchala mato jami necha metr?

1. Adrasga mos sonlar 5 va 3, EKUK — 15. $3:5 = 9:15$, $3:2 = 15:10$.
2. a) Atlas : adras : shoyi $= 9:15:10$.
3. Atlas adrasdan $15 - 9 = 6$ bo'lak qisqa: 6 bo'lak = 18 m.
4. 1 bo'lak $= 18:6 = 3$ m.
5. b) Adras shoyidan $15 - 10 = 5$ bo'lak uzun: $5 \cdot 3 = 15$ m.
6. c) Jami $9 + 15 + 10 = 34$ bo'lak: $34 \cdot 3 = 102$ m.

3-MISOL · KVADRAT SOHALARI

Kvadrat diagonal bilan teng ikki yarmiga bo'lingan, har yarmi yana ikki sohaga ajratilgan. $K:L = 1:3$, $K:N = 2:3$. Kvadratning yuzi 80 cm². a) $K:L:M$ ni toping. b) M sohaning yuzini toping.



1. K ga mos sonlar 1 va 2, EKUK — 2. $1:3 = 2:6$. $K:N = 2:3$ o'zgarmaydi — unda K allaqachon 2. Demak $K = 2$, $L = 6$, $N = 3$ bo'lak.
2. Diagonal kvadratni teng ikkiga bo'ladi: $K + L = M + N$. *(Ikki yarmining yuzi teng.)*
3. $2 + 6 = M + 3$, $M = 8 - 3 = 5$ bo'lak. a) $K:L:M = 2:6:5$.
4. Kvadrat $2 + 6 + 5 + 3 = 16$ bo'lak = 80 cm², 1 bo'lak $= 80:16 = 5$ cm².
5. b) $M = 5 \cdot 5 = 25$ cm².



Bu xatolarni men ham qilganman!

DIQQAT!

1. Umumiy miqdorning hadlari tenglashmaguncha uch hadli nisbat yozilmaydi. $A : B = 7 : 2, B : C = 3 : 5$:
   ✗ $A : B : C = 7 : 2 : 5$ → ✓ $21 : 6$ va $6 : 10, A : B : C = 21 : 6 : 10$
2. Nisbatning ikkala hadini ko'paytiring, faqat bittasini emas. $K : L = 1 : 3, L : M = 2 : 3$:
   ✗ $K : L : M = 1 : 6 : 9$ → ✓ $1 : 3 = 2 : 6, 2 : 3 = 6 : 9, K : L : M = 2 : 6 : 9$
3. Kasrdagi tartibni almashtirib yubormang. «B soni A sonining $\frac{2}{3}$ qismi»:
   ✗ $A : B = 2 : 3$ → ✓ $B : A = 2 : 3$, ya'ni $A : B = 3 : 2$
4. Javobni savoldagi tartibda yozing. Qalam : ruchka $= 4 : 5$, qalam : chizg'ich $= 6 : 11$; so'ralgan: ruchka : qalam : chizg'ich.
   ✗ $12 : 15 : 22$ → ✓ qalam 12, ruchka 15, chizg'ich 22 bo'lak: $15 : 12 : 22$

3-QISM

Mustahkamlash



Chizmali model chizsangiz, masala tezroq yechiladi.

1. Gul do'konida oq : pushti atirgullar $= 3 : 5$, pushti : qizil atirgullar $= 2 : 3$. Oq : pushti : qizil nisbatini toping. Chizmali model chizing.

2. Uch hadli nisbat ko'rinishida yozing: a) $P : Q = 7 : 6$, $Q : R = 4 : 3$; b) $M : N = 3 : 2$, $N : K = 8 : 5$; c) $E : F = 5 : 4$, $F : G = 4 : 9$.

3. $A : B : C$ ni toping: a) $A : B = 4 : 3$, $A : C = 10 : 7$; b) $A : C = 3 : 8$, $B : C = 7 : 6$.

4. $A : B : C$ ni toping: a) B soni A sonining $\frac{3}{4}$ qismi, C soni A dan 2 marta ko'p; b) A soni B sonining $\frac{2}{5}$ qismi, C soni B dan 3 marta kam.

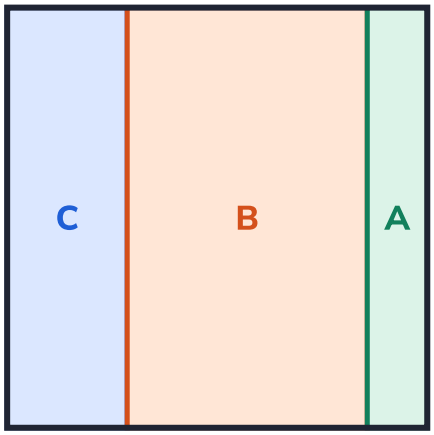
5. Akvariumda oltin : kumush baliqchalar $= 3 : 4$, kumush : qora baliqchalar $= 6 : 5$. a) (Oltin + kumush) : qora nisbatini toping. b) Qora baliqchalar hammasining qanday qismi?

6. $X : Y = 5 : 4$ va $Y : Z = 6 : 13$. Nilufar $X : Y : Z = 15 : 12 : 26$ deb topdi, Temur esa $10 : 8 : 13$ deb. Kim to'g'ri topgan? Javoblarni berilgan nisbatlar bilan solishtiring.

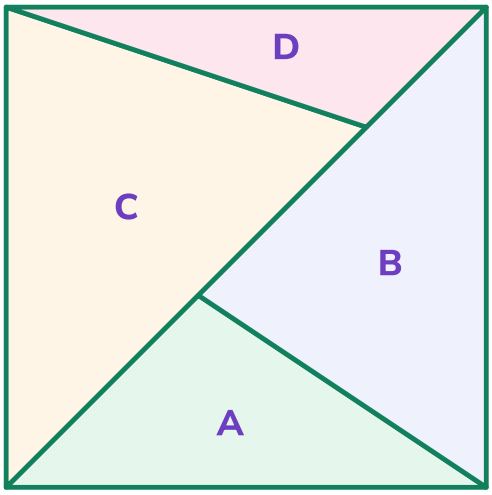
7. Kutubxona javonida ertak : she'r kitoblari $= 5 : 2$. She'r kitoblari hikoya kitoblarining $\frac{2}{7}$ qismi. Javonda jami 280 ta kitob bor. a) Ertak, she'r va hikoya kitoblari nisbatini toping. b) Hikoya kitoblari nechta?

8. Gulzorda qizil : oq atirgullar $= 7 : 2$, qizil : sariq atirgullar $= 3 : 5$. Oq va sariq atirgullar birgalikda 205 ta. a) Oq : qizil : sariq nisbatini toping. b) Qizil atirgullar nechta?

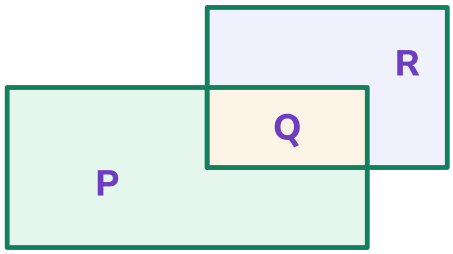
9. Kvadrat A, B, C sohalarga bo'lingan. C sohaning yuzi A sohanikidan 2 marta katta, $B : C = 2 : 1$. a) $A : B : C$ ni toping. b) B soha A sohadan 21 cm² katta bo'lsa, kvadratning yuzi qancha?



10. Kvadrat diagonal bilan teng ikki yarmiga bo'lingan, har yarmi yana ikki sohaga ajratilgan. $B : A = 3 : 2$, $C : B = 5 : 4$, D sohaning yuzi 15 cm². a) D soha necha bo'lak? b) Kvadratning yuzini toping.



11. Chapdagi va o'ngdagi to'g'ri to'rtburchaklar ustma-ust tushgan. Q qism chapdagisining $\frac{2}{9}$ qismi, o'ngdagisining $\frac{1}{3}$ qismi. a) $Q : P : R$ ni toping. b) Chapdagi va o'ngdagi to'rtburchaklar yuzlari nisbati qanday? *(Maslahat: Q ni 2 bo'lak deb oling.)*



4-QISM

Mustaqil ish

1. Avtobekatda avtobuslar : taksilar $= 2 : 7$, avtobuslar : mikroavtobuslar $= 3 : 8$. Avtobuslar, taksilar va mikroavtobuslar sonlari nisbatini toping.
2. Qisqarmas ko'rinishda yozing: a) $K : L = 6 : 4$, $L : M = 10 : 15$; b) $X : Y = 3 : 4$, $Y : Z = 6 : 10$.

3. Uch idishda suv bor. 1-idishdagi suv 2-idishdagining $\frac{3}{4}$ qismi. 3-idishdagi suv 2-idishdagidan 3 marta kam. 1-idishda 3-idishdagidan 20 litr ko'p suv bor. Uchala idishda jami necha litr suv bor?

4. Uch quti olmaning massalari: birinchi : ikkinchi $= 9:4$, ikkinchi : uchinchi $= 6:5$. Uchala qutida jami 98 kg olma bor. Eng og'ir quti eng yengilidan necha kg og'ir?

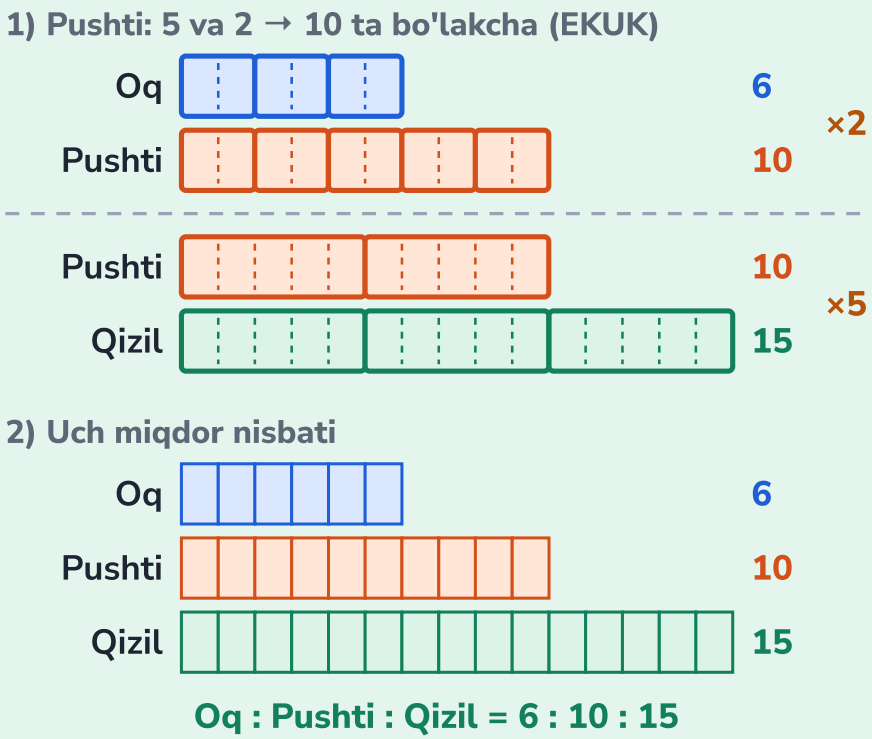
5. **Xatoni toping.** Bobur $A:B=4:3$ va $B:C=2:7$ dan «$A:B:C=4:3:7$» deb yozdi. To'g'ri javobni yozing.

5-QISM

Javoblar

Mustahkamlash

1-masala



Pushti birinchi nisbatda 5, ikkinchisida 2 bo'lak; EKUK 10. $3:5=6:10$, $2:3=10:15$. Oq : pushti : qizil $=6:10:15$.

2-masala

a) 6 va 4 ning EKUKi 12. $7:6=14:12$, $4:3=12:9$, $P:Q:R=14:12:9$.
b) 2 va 8 ning EKUKi 8, faqat birinchisi ×4. $3:2=12:8$, $M:N:K=12:8:5$.
c) F ikkala nisbatda 4 bo'lak — hech narsa ko'paytirilmaydi: $E:F:G=5:4:9$.

3-masala

a) Ikkala nisbatda A bor: 4 va 10, EKUK 20. $A:B=20:15$, $A:C=20:14$. $A:B:C=20:15:14$. b) Bu safar C ikkalasida oxirida: 8 va 6, EKUK 24. $A:C=9:24$, $B:C=28:24$. $A:B:C=9:28:24$.

4-masala

a) $A:B=4:3$, $A:C=1:2=4:8$. $A:B:C=4:3:8$. b) $A:B=2:5$, $C:B=1:3$. 5 va 3 ning EKUKi 15. $A:B=6:15$, $C:B=5:15$. $A:B:C=6:15:5$.

5-masala

Kumush baliqchalarga mos sonlar 4 va 6, EKUK 12. $3:4=9:12$, $6:5=12:10$. Oltin : kumush : qora $=9:12:10$. a) $(9+12):10=21:10$. b) Hammasi $9+12+10=31$ bo'lak, qora — 10 bo'lak: $\frac{10}{31}$ qismi.

6-masala

Nilufar: $15:12=5:4$ ✓, $12:26=6:13$ ✓ — to'g'ri. Temur: $10:8=5:4$ ✓, lekin $8:13$ nisbat $6:13$ ga teng emas — noto'g'ri (Y ning hadlari tenglashtirilmagan).

7-masala

She'r : hikoya $=2:7$. She'r ikkala nisbatda 2 bo'lak. a) Ertak : she'r : hikoya $=5:2:7$. b) Jami $5+2+7=14$ bo'lak = 280, 1 bo'lak $=280:14=20$. Hikoya kitoblari $7\cdot 20=140$ ta.

8-masala

a) Qizilga mos sonlar 7 va 3, EKUK 21. $7:2=21:6$, $3:5=21:35$. Oq : qizil : sariq $=6:21:35$. b) Oq va sariq $6+35=41$ bo'lak = 205, 1 bo'lak $=205:41=5$. Qizil $21\cdot 5=105$ ta.

9-masala

a) $A:C=1:2$. C ga mos sonlar 2 va 1, EKUK 2. $B:C=2:1=4:2$. $A:B:C=1:4:2$. b) $4-1=3$ bo'lak = 21 cm², 1 bo'lak $=21:3=7$ cm². Kvadrat $1+4+2=7$ bo'lak: $7\cdot 7=49$ cm².

10-masala

a) $A:B=2:3$, $B:C=4:5$. B ga mos sonlar 3 va 4, EKUK 12. $A:B=8:12$, $B:C=12:15$, demak $A:B:C=8:12:15$. $A+B=C+D$, shuning uchun $D=8+12-15=5$ bo'lak. b) 5 bo'lak = 15 cm², 1 bo'lak $=15:5=3$ cm². Kvadrat $8+12+15+5=40$ bo'lak: $40\cdot 3=120$ cm².

11-masala

a) Q — 2 bo'lak. U chapdagisining $\frac{2}{9}$ qismi, demak chapdagisi 9 bo'lak. U o'ngdagisining $\frac{1}{3}$ qismi, demak o'ngdagisi $2 \cdot 3 = 6$ bo'lak. $P = 9 - 2 = 7$, $R = 6 - 2 = 4$. $Q : P : R = 2 : 7 : 4$. b) $9 : 6 = 3 : 2$.

Mustaqil ish

1-masala

Avtobuslarga mos sonlar 2 va 3, EKUK 6. $2 : 7 = 6 : 21$, $3 : 8 = 6 : 16$. Javob: $6 : 21 : 16$.

2-masala

a) Avval qisqartiramiz: $6 : 4 = 3 : 2$, $10 : 15 = 2 : 3$. L ikkalasida 2 bo'lak: $K : L : M = 3 : 2 : 3$. b) $6 : 10 = 3 : 5$. 4 va 3 ning EKUKi 12. $3 : 4 = 9 : 12$, $3 : 5 = 12 : 20$. $X : Y : Z = 9 : 12 : 20$.

3-masala

1-idish : 2-idish $= 3 : 4$, 3-idish : 2-idish $= 1 : 3$. 4 va 3 ning EKUKi 12. $3 : 4 = 9 : 12$, $1 : 3 = 4 : 12$. Idishlar $9 : 12 : 4$. 1-idish 3-idishdan $9 - 4 = 5$ bo'lak ko'p: 5 bo'lak = 20 l, 1 bo'lak $= 20 : 5 = 4$ l. Jami $9 + 12 + 4 = 25$ bo'lak: $25 \cdot 4 = 100$ l.

4-masala

Ikkinchi qutiga mos sonlar 4 va 6, EKUK 12. $9 : 4 = 27 : 12$, $6 : 5 = 12 : 10$. Qutilar $27 : 12 : 10$, jami 49 bo'lak = 98 kg, 1 bo'lak $= 98 : 49 = 2$ kg. Eng og'iri 27 bo'lak, eng yengili 10 bo'lak: farq $17 \cdot 2 = 34$ kg.

5-masala

B birinchi nisbatda 3 bo'lak, ikkinchisida 2 bo'lak — tenglashtirilmagan. 3 va 2 ning EKUKi 6. $4 : 3 = 8 : 6$, $2 : 7 = 6 : 21$. To'g'ri javob: $A : B : C = 8 : 6 : 21$.